

Fakhreddin (Amir) Amirhosseini

Mechatronic Systems Engineer

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PROFESSIONAL SUMMARY

Mechatronics engineer with near a decade of experience in designing and developing production-grade hardware and software. Recognized for a wide cross-disciplinary technical skills, strong problem-solving ability, with the project ownership and self-management ability to develop engineering solutions for real industrial problems.

WORK EXPERIENCE

Automation Engineer Intern

Jan 2026 – Now

Kardium – *Burnaby, BC*

- Kardium is a medical device company. I developed and maintained software and hardware for automation jigs for manufacturing a catheter for treating a heart condition called atrial fibrillation.
- Developed and contributed to C# and Python codebases, including UIs, vision, hardware control, and data analyzers, following SOLID principles with complete unit testing, and Git and TeamCity version control.
- Designed and modified high-frequency, high-voltage electrical schematics and PCBs in Altium Designer for automation jigs, following full product lifecycle processes within a PLM system called Windchill.
- Cross-team and cross-disciplinary collaboration with mechanical, manufacturing, quality control, and production teams, following strict medical development guidelines, codes, and standards.

Mitacs Intern and Graduate Research Assistant

Sep 2023 – Jul 2025

Aluminate Inc. and Intelligent Sensing Lab at Simon Fraser University (SFU) – *Surrey, BC*

- Aluminate is a tech developer for the aluminum smelting industry. I designed high-precision sensors for two long-standing technical problems: measurement of molten bath level and its chemical composition.
- I designed 3 novel sensing techniques using resonant capacitive and eddy-current sensors to be used in the harsh smelting environment, including 1000° C temperature, corrosive chemicals, and strong EMI.
- Published the designs, experimental measurement results, and data validation technique in a provisional US patent, an award-winning conference paper, and a featured paper in IEEE Sensors Journal.
- Modified R&D timeline and milestones according to customers' needs and pinpoints and coordinated with multiple suppliers and vendors in Canada, China, and the US to build prototypes within R&D budget.
- Used advanced electrical measurement tools (e.g., network and impedance analyzers), and COMSOL simulation to model and characterize the sensors' behavior, optimize the design, and validate data.
- Organized field test logistics including coordination with a group of 8 engineers and technicians, developing contingency plans, and on-site measurement in a designated aluminum smelting test facility.

Product Development Lead

Jun 2017 – Sep 2023

Tart Robotics – Toronto, ON

- Tart Robotics is an STEM education platform manufacturer. I designed 3 robotic platforms and helped take them from concept to successful products, with over 3,000 units sold globally. [See products](#).
- Developed software support including Arduino libraries, MATLAB Simulink toolkits, Python packages, and graphical programming platforms in Node JS and Flutter for controlling the hardware through BLE.
- Designed PCB enclosures, motor housings, and other snap-fit reconfigurable plastic components with high tolerances for mass production through injection molding in SolidWorks.
- Designed proprietary electronics for an in-house low-power ultrasonic sensor and a leak-free soft power switch for circuits with non-removable batteries, resolving manufacturing and stocking challenges.
- Mentored and supported 6 interns during internship programs, providing exposure to mechanical design, software, hardware, and firmware development, resulting in multiple recruitments for the company.

Research Assistant

Sep 2020 – Jul 2023

Kharazmi University – Tehran, Iran

- As a robotic and AI researcher, I investigated motion planning algorithms for autonomous mobile robots and applied deep learning in robotics and hardware for custom use cases in robotics.
- Created LPM-Net: a predictive Deep Learning model for motion planning, with 50% less computational load than MPC, packaged open source for ROS and published in a journal article. [Journal link](#), [Github link](#)
- Built machine vision pipelines using YOLO and SSD models with MobileNet frameworks for ROS, implemented on Raspberry Pi for object tracking. [Github link](#).

Research Assistant

Sep 2018 – Sep 2020

Microfluidics Lab at Qazvin International University – Qazvin, Iran

- Worked on B.Sc. thesis to design and fabricate a microfluidic lab-on-a-chip for separating circulating tumor cells from blood sample using PDMS fabrication and photolithography.

Research Assistant

Sep 2017 – Sep 2018

IoT Lab at Qazvin International University - Qazvin, Iran

- Designed a gate-level address-decoder for ultra-low-power self-sufficient IoT devices, significantly reducing power from $\sim 100 \mu W$ to 68 nW, resulting in a journal publication. [Journal link](#).

EDUCATION**Simon Fraser University - Surrey, BC**

Sep 2023 – Jan 2026

M.A.Sc. – Mechatronic Systems Engineering

Simon Fraser University Beedie School of Business – Burnaby, BC

Sep 2024 – Apr 2025

Graduate Certificate – i2I, Science and Technology Commercialization

Kharazmi University - Tehran, Iran

Sep 2020 – Jun 2023

M.Sc. – Artificial Intelligence

Qazvin International University - Qazvin, Iran

Sep 2016 – Jun 2020

B.Sc. – Electrical and Control Engineering

AWARDS AND RECOGNITIONS

Nominated for the Governor General's Academic Medal.	Jan 2026
Received Outstanding Lecture Award at IEEE Sensors conference among 400+ accepted works.	Oct 2025
Graduate Fellowship recipient (3x)	2023 - 2025
Recipient of full scholarship for all education levels.	2016 - 2026
(Fun fact!) Featured multiple times on Hackster.com and Instructable.com .	2017 - 2023

LANGUAGE

- English – professional proficiency (CELP: 11/12, IELTS: 8/9)
- Persian (Farsi) – native

SKILL

- C#, .Net, C/C++, Python, JavaScript, Git. Develop professional and robust code, following SOLID principles and complete unit testing. Proficiency with Git working on large, multi-authored code bases.
- SolidWorks and Altium Designer. Experienced with PLM platforms like Windchill. PCB design for high-frequency and high-voltage applications.
- Deep Learning, TensorFlow, Machine Vision with custom applications.
- R&D, Scientific Problem Solving, COMSOL RF and AC/DC.
- Sensors: IMU, Camera, Sonar, Inductive, Capacitive.
- ROS, Robot Manipulation, and Motion Planning.